

ECOLOGICAL REPORT

Sites: TA 00001, TA 00002

PRELIMINARY ECOLOGICAL APPRAISAL

Preliminary Ecological Appraisal (PEA) -
Test apro

Prepared For	Client
Site Reference	TA 00001, TA 00002
Report Version	Version 1
Date	11/5/2026

TABLE OF CONTENTS

1.	Introduction	3
2.	Methodology	4
3.	Results	5
4.	Ecological Constraints	6
5.	Discussion & Conclusions	7
6.	Appendices	8

APPENDICES

Appendix A	Habitat Map
Appendix B	Designated Sites
Appendix C	Species List
Appendix D	Aquatic Features
Appendix E	Site Photographs
Appendix F	Habitat Data
Appendix G	Survey Datasheets
Appendix H	Legislation References

1. Introduction

This Preliminary Ecological Appraisal (PEA) has been prepared to characterise the baseline ecological conditions and identify constraints and opportunities for a proposed development at **Test apro**(Site Code: TA-2026-421), located in the townland of **Shanettra, County Mayo**, Connacht province (Grid Reference: E 062 293). The assessment is conducted in accordance with **Conseil International pour l'Exploration de la Mer (CIEEM) Guidelines for Ecological Impact Assessment** and provides a foundation for environmental decision-making, including preliminary screening for Appropriate Assessment (AA) under the **EU Habitats Directive (92/43/EEC)** and **Birds Directive (2009/147/EC)**.

Purpose and Scope of Assessment

The PEA encompasses desk-based habitat and species data collation, field survey validation, and identification of potential ecological receptors within and immediately adjacent to the development boundary. The assessment is designed to:

- characterise existing habitats using the **FOSSITT habitat classification**(Fossitt, 2000) and the National Land Cover Map (NLC) 2018
- identify European designated sites (Special Areas of Conservation [SACs], Special Protection Areas [SPAs], proposed Natural Heritage Areas [pNHAs] and Natural Heritage Areas [NHAs]) within a search radius and assess potential for connectivity
- screen for Appropriate Assessment requirements under the European Communities (Environmental Assessment of Certain Plans and Programmes) Regulations 2004 (as amended)
- record protected and priority species and invasive alien species present or likely within the study area
- establish baseline conditions against which potential project impacts can be assessed in subsequent detailed Environmental Impact Assessment (if required)

Field surveys undertaken between April and May 2026 include habitat mapping, vegetation relevés, aquatic feature assessment and general ecological walkovers (with a final walkover survey in progress). A total of **2,963.32 hectares** of habitat across 31 distinct types has been recorded within the broader survey extent.

Relevant Legislation and Policy Framework

This assessment operates within a hierarchical regulatory framework requiring strict adherence to:

- **Wildlife Acts 1976–2021**: Irish primary legislation protecting fauna and flora, including species schedules and habitat provisions; compliance is mandatory for all works that may disturb protected species or habitats
- **EU Habitats Directive (92/43/EEC)** and **Birds Directive (2009/147/EC)**: These establish legal obligations to assess impacts on European sites and their qualifying habitats and species; projects with potential for significant effects must be subject to Appropriate Assessment
- **European Communities (Environmental Assessment of Certain Plans and Programmes) Regulations 2011**(as amended): Requires AA screening for plans and programmes (e.g., forestry, peatland, water) with potential hydrological or functional connectivity to Natura 2000 sites
- **Water Framework Directive (2000/60/EC)**: Projects with surface or groundwater connections must be assessed against WFD objectives; deterioration of water body status is prohibited

- **Planning and Development Acts 2000–2023**(Ireland): Environmental and ecological assessment is embedded within statutory development control; many Local Authorities require PEAs and Natura Impact Statements as planning conditions

Geographical and Ecological Context

The site is located within the **Blacksod-Broadhavencatchment** (EPA Code 33; River Basin District: Irish Eastern and Western), a Water Framework Directive management unit that drains to the **Mullet/Blacksod Bay Complex SAC (IE0000470)**. The **Owenmore (Mayo) River** is recorded approximately **0.51 km** from the site, representing a potential hydrological receptor. Within a 1 km buffer zone, the landscape is dominated by:

- **Peatland habitats:** Upland blanket bog (PB2, 685.0 ha), cutover bog (PB4, 153.65 ha)
- **Heathland:** Dry siliceous heath (HH1, 34.55 ha) and wet heath (HH3, 40.1 ha)
- **Woodland and scrub:** Immature woodland (WS2, 1,507.9 ha), conifer plantation (WD3, 413.7 ha), semi-natural woodland (WN, 17.16 ha) and scrub (WS1, 31.89 ha)

Three European designated sites occur within 3.5 km: **Glenamoy Bog Complex SAC (1.67 km)**, **Bellacorick Bog Complex SAC (3.52 km)**— both harbouring Annex I peatland and heath habitats of high conservation value — and multiple NHAs/pNHAs supporting similar peatland communities.

Report Structure

This document is organised as follows:

- **Introduction**(this section): Project context, regulatory framework and geographical overview
- **Methodology:** Survey methods, effort, timing and limitations
- **Baseline Conditions:** Desktop and field habitat assessment, species records, and designated site characterisation
- **Ecological Receptors and Constraints:** Summary of sensitive habitats, protected species and water environment
- **Data Gaps and Further Assessment:** Identification of information deficits requiring targeted field surveys prior to design/consent
- **Conclusions and Recommendations:** Preliminary Appropriate Assessment screening outcome and outline mitigation requirements

2. Methodology

Methodology

Desk Study

A comprehensive desk-based review was undertaken to collate existing ecological, hydrological and statutory data relevant to the site and its environs. Data sources consulted included:

Designated Sites & Spatial Layers:

- National Parks and Wildlife Service (NPWS) database — searched for Special Areas of Conservation (SAC), Special Protection Areas (SPA), Natural Heritage Areas (NHA) and proposed NHA (pNHA) designations within a 5 km radius of the development site. Five designated sites were identified within 3.5 km, including Bellacorick Bog Complex SAC (001922) at 3.52 km, and Glenamoy Bog Complex SAC (000500) at 1.67 km, alongside multiple peatland-designated NHAs.
- Environmental Protection Agency (EPA) catchment and water body datasets — identified two Blacksod-Broadhaven catchment records (Catchment Code 33, River Basin District: IEWE) and confirmed hydrological connectivity via the Owenmore (Mayo) River (0.51 km downstream). Water Framework Directive (WFD) status data for individual river bodies was not made available in the provided source material and should be obtained directly from EPA WFD portal during Appropriate Assessment Screening.
- National Land Cover Map (NLC) 2018 — habitat data were extracted using FOSSITT (2000) classification, yielding 31 distinct habitat polygons across the survey area totalling 2,963.32 hectares. Within the development boundary, three habitat types were identified: WS2 Immature woodland (772.59 ha), PB4 Cutover bog (12.32 ha) and PB2 Upland blanket bog (5.17 ha). Within the 1 km buffer zone (outside boundary), 11 additional habitat types were recorded, dominated by WS2 Immature woodland (735.31 ha), WD3 Conifer plantation (413.70 ha combined), and peatland/heath mosaics (PB2, PB4, HH1, HH3).

Protected and Invasive Species:

- NBDC/GBIF databases and NPWS species records — searches for protected species records within 5 km yielded 23 records of Goldcrest (*Regulus regulus*), a Birds of Conservation Concern (BoCCI) Amber-listed species protected under the Wildlife Acts 1976–2021. No detailed species occurrence data (counts, locations, dates) were provided by these sources for the development site itself, representing a key data gap addressed through field survey recommendations below.
- Invasive Species Ireland and IUCN Invasive Species Specialist Group — desktop review identified seven invasive/non-native species of concern within the broader survey area, including three high-risk taxa listed under EU Regulation 1143/2014 (*Gunnera tinctoria*, *Rhododendron ponticum*, Himalayan knotweed *Persicaria wallichii*) and four medium-risk species (*Sycamore Acer pseudoplatanus*, Jenkins' Spile Snail *Potamopyrgus antipodarum*, Rabbit *Oryctolagus cuniculus*, Black Currant *Ribes nigrum*). Presence on the development site has not yet been confirmed through field survey.

Field Survey

A total of six ecological survey visits were conducted between 22 April and 9 May 2026. Survey effort comprised habitat mapping, relevé vegetation surveys, aquatic sampling and general walkovers as summarised below.

Survey Dates and Weather Conditions:

All surveys except the walkover were completed on **22 April 2026** under favourable conditions: visibility good, precipitation none, air temperature 15°C, north wind at 100 km/h, cloud cover 50%. The ongoing walkover survey commenced on **9 May 2026**.

Habitat Mapping Survey (22 April 2026)

A targeted habitat mapping survey was completed, recording habitat extent across a 50 hectare trial area. Habitat classification followed Fossitt (2000), with recording of the following parameters: habitat type code, total area (50 ha), dominant species composition (recorded as categorical description), structural variables (shrub cover 50%, canopy cover 50%, ground cover 50%), condition assessment (classified as 'good'), and vegetation structure observations. Visibility was good throughout the survey period, with no precipitation and steady north wind (100 km/h). The habitat was recorded under FOSSITT as a test code with 50% shrub and canopy coverage, 15°C ambient temperature and 50% cloud cover. **Limitation:** This represents a small pilot survey and does not cover the full extent of the development site; comprehensive habitat mapping across all site polygons is required during the May–August peak season to verify NLC 2018 designations.

Relevé Vegetation Surveys (22 April 2026)

Two floristic relevé surveys were completed at co-located grid references (53.3498°N, 6.2603°W, accuracy ±5 m) on peat soils. Both surveys recorded identical vegetation structure and species composition under FOSSITT habitat type FL1.

Parameter	**REL 101 & REL 102**
**Total vegetation...	90%
**Dominant grami...	40% (median height 20 cm, max 40 cm)
Forbs	25% (median 15 cm, max 30 cm)
**Bryophytes (mo...	15% (max 5 cm)
Shrubs	5% (max height 150 cm)
Trees	5% (max height 8 m)
Bare soil & rock	3% and 1% respectively
Litter & water	5% litter, 1% open water
Soil type	Peat, firm stability
Slope & aspect	5° slope, south-facing
**Species record...	* <i>Agrostis capillaris</i> * (Common Bent, DOMIN 6, 30%); * <i>Anthoxanthum odoratum</i> * (Sweet Ve...

Recorder:test | **Accuracy:**±5 m GPS

These relevés document a species-poor, grass-dominated community on peaty substrate, consistent with upland modified grassland or heath transition. **Limitation:** Only two relevés were completed at a single grid reference on a single date in April; a full spring–summer botanical survey (May–August) across multiple habitat units (particularly PB2, HH1, HH3, GS4 polygons) is required to characterise Annex I habitat potential and document full species richness.

Aquatic Survey (22 April 2026)

One aquatic survey was completed on a river water body (average depth 2 m, channel width 50 m, substrate bedrock). In-situ measurements included:

Parameter	**Value**
Water pH	7 (neutral)
Turbidity	Clear
Dissolved oxygen	50 mg/L
Conductivity	50 µS/cm
Temperature	15°C
Sampling method	Kick sample
Aquatic species recorded	Test Species Recorded (nomenclature/counts not specified)

Visibility was good; sampling occurred under north wind (100 km/h), with no precipitation. **Limitations:** Species nomenclature and quantitative data are incomplete; no macro-invertebrate identification list or ecological classification (e.g., WFD or Q-value assessment) was provided; sampling represents one site visit and does not characterise the full watercourse length or seasonal variation in water quality.

General Walkover Survey (9 May 2026)

A general walkover survey was initiated on 9 May 2026 and remains in progress. This survey is designed to verify habitat extent, check for protected species activity (birds, bats, otters) and assess invasive species distribution across accessible site areas.

Survey Limitations

The surveys completed to date are subject to the following constraints:

- **Seasonal timing:** April surveys fall at the margin of the optimal

3. Results

- Results

3.1 Designated Sites

The desk-based assessment has identified six designated sites within 2 km of the project boundary, comprising two European sites (SACs) and four National/proposed National Heritage Areas (NHAs/pNHAs), all within the Bellacorick Bog Complex and surrounding peatland landscape (Table 3.1). The closest designated site is Inagh Bog NHA, located 1.52 km from the site boundary. Of primary conservation significance are **Glenamoy Bog Complex SAC** (1.67 km) and **Bellacorick Bog Complex** (pNHA at 1.67 km and SAC at 3.52 km), which are regionally important peatland mosaic systems with multiple Annex I habitat and species interests.

Designat...	Type	Distan...	Qualifying Interests	Conser
Glenamo...	SAC	1.67	Not fully specified in dataset	—
Bellacori...	pNHA	1.67	Natural dystrophic lakes (3160), Wet heaths (4010), Blanket ...	0 Favourable,...

Bellacori...	SAC	3.52	Natural dystrophic lakes (3160), Wet heaths (4010), Blanket ...	0 Favourable,...
Inagh Bog	NHA	1.52	Not specified	—
Ummera...	NHA	1.68	Not specified	—

Detailed Conservation Assessment

Bellacorick Bog Complex (SAC/pNHA, 1.67–3.52 km) is a large, intact peatland mosaic (9586 ha) of high conservation significance comprising blanket bogs, wet heaths, Rhynchosporion depression fens, natural dystrophic lakes and alkaline fens. The site supports Annex II species including *Vertigo geyeri* (Hairy Snail) and *Saxifraga hirculus* (Yellow Marsh-saxifrage), and maintains critical hydrological and carbon storage functions. Conservation status across the five qualifying features is mixed, with four habitats in poor ('bad') condition, reflecting widespread pressures from drainage, peat extraction, forestry encroachment, overgrazing, burning and peat cutting. Climate change, agricultural intensification and nutrient enrichment pose emerging threats to the site's ecological integrity.

The **Glenamoy Bog Complex SAC** (1.67 km) is similarly a peatland-dominated system of regional importance, though detailed qualifying interest data were not available in the provided datasets.

Hydrological and Functional Connectivity

The site lies within the **Blacksod-Broadhaven catchment** (EPA Code 33, River Basin District: IEWE), a Water Framework Directive (WFD) management unit linked downstream to the **Mullet/Blacksod Bay Complex SAC** (IE0000470), a coastal site qualifying for mudflats, sandflats and shallow bays (Annex I habitats 1140, 1160) and supporting otter (*Lutra lutra*, Annex II species). The **Owenmore (Mayo) River** is recorded 0.51 km from the site boundary, suggesting a plausible surface-water hydrological connection. Any groundwater or surface drainage from the project lands has the potential to affect WFD status and the conservation interests of downstream European sites. Accordingly, **Appropriate Assessment Screening is required**, and precautionary hydrological and water-quality protections are recommended (see Section 5 — Recommendations).

Additional NHAs (Inagh Bog, Ummerantarry Bog) at 1.52–1.68 km are peatland-dominated heritage sites of regional value, though detailed ecological data were unavailable for full assessment in this preliminary appraisal.

3.2 Habitats

The site encompasses a diverse mosaic of habitats across 2963.32 hectares, dominated by immature woodland, peatland, forestry and heath (Table 3.2). Habitats were classified using the **FOSSITT 2000** system (Fossitt, 2000) and derived primarily from the National Land Cover Map 2018 (Tailte Éireann).

Habitat Summary

FOSSIT...	Habitat...	Total A...	Condition	Annex I Potential	Notes
PB2	Upland ...	684.95	Presume...	7130 Blanket bogs ...	Dominant; 22.9% of total area. Peat depth, wa...

WS2	Immatu...	1507.90	Good (pr...	Possible 9120 Atla...	Largest habitat; 50.4% of total area. Transition...
WD3	Conifer ...	413.70	Not asse...	None (non-native)	13.9% of total area. Two woodland blocks (145...
HH3	Wet he...	40.10	Not asse...	4010 Northern Atla...	1.4% of total area. Two blocks (2.47 ha, 37.63 ...
HH1	Dry silic...	34.55	Not asse...	4030 European dry...	1.2% of total area. Two blocks (4.55 ha, 30.00 ...
PB4	Cutover...	153.65	Poor to m...	7130 Blanket bogs ...	5.2% of total area. Three blocks ranging 12.32...
WS1	Scrub	31.89	Not asse...	None as Annex I	1.1% of total area. Two blocks (0.69 ha, 31.20 ...
GS4	Wet gra...	38.66	Not asse...	6410 Molinia mead...	1.3% of total area. Two blocks (19.19 ha, 19.4...
FW	Waterc...	6.27	Not asse...	3260 Fluvial water ...	0.2% of total area. Requires assessment for hy...
BL3	Building...	28.07	Not asse...	None	0.9% of total area. Four separate features (0.5...
ER	Expose...	2.61	Not asse...	None (unless coast...	0.1% of total area. Two blocks (1.18 ha, 1.43 h...
WN	Semi-n...	17.16	Presume...	9120 Atlantic acido...	0.6% of total area. Two blocks (0.16 ha, 17.00 ...
GA1	Improv...	0.56	Good	None	<0.1% of total area. Low conservation sensitivi...
HD1	Dense ...	0.54	Fair	None (but indicator...	<0.1% of total area. Requires nesting bird che...
WL2	Treelines	0.16	Not asse...	None as primary	<0.1% of total area. Medium sensitivity; bat P...
WL1	Hedger...	1.55	Not asse...	None as standalone	<0.1% of total area. Assess connectivity and n...

Key Habitat Findings

Peatland Complex (PB2, PB4):The site is dominated by blanket bog (PB2: 684.95 ha) and cutover bog (PB4: 153.65 ha), collectively representing 27.1% of the total area and of high conservation potential. Upland blanket bog (PB2) is an Annex I habitat (code 7130, *Blanket bogs* with priority status if actively forming). Field verification is essential to confirm peat depth (>30–50 cm threshold), degree of drainage/erosion and hydrological integrity. Cutover bog (PB4) represents heavily degraded peatland with poor current conservation status but potential for restoration depending on peat stability and drainage status.

Heathland Mosaic (HH1, HH3):Wet and dry siliceous heaths collectively occupy 74.65 ha (2.6% of area). Both are potential Annex I habitats: wet heath (4010 *Northern Atlantic wet heaths with Erica tetralix*) and dry heath (4030 *European dry heaths*). Field botanical surveys in peak season (May–August) are required to assess vegetation composition, cover of diagnostic species (*Erica tetralix*, *Calluna vulgaris*, *Erica cinerea*), hydrological status and conservation condition.

Woodland and Plantation:Immature woodland (WS2: 1507.90 ha, 50.4%) and conifer plantation (WD3: 413.70 ha, 13.9%) together form extensive tree cover. Immature woodland represents successional vegetation on formerly open ground, potentially transitional towards semi-natural communities; Phase 1 habitat mapping is required to distinguish between native broadleaved woodland (potential Annex I 9120) and planted or volunteer colonisation. Conifer plantation is of lower conservation value but may support bat roosts and nesting birds; pre-felling PRF and breeding bird surveys are essential. Semi-natural woodland (WN: 17.16 ha) is present in two small blocks and warrants botanical and bat surveys to establish conservation significance.

Grassland and Wetland:Wet grassland (GS4: 38.66 ha) occurs in two blocks and may qualify as Annex I molinia meadows (6410) if floristically diverse and nutrient-poor; botanical survey in June–August is recommended. Watercourses (FW: 6.27 ha) require hydrological and ecological assessment, including investigation of riparian vegetation quality and potential salmonid or otter use.

Habitat Condition and Connectivity

Field surveys on **22 April 2026**(habitat mapping survey) confirmed visibility as good and general habitat condition as good for the mapped sample. However, this early-season survey does not provide comprehensive botanical data sufficient to assess Annex I status or conservation condition of peatland and heathland habitats; **May–August surveys are strongly recommended**to capture peak floristic diversity and to verify habitat quality against EU Habitats Directive criteria.

The habitat mosaic exhibits high ecological connectivity, particularly between peatland and heath complexes (PB2–HH1/HH3 transitions) and between woodland blocks (WS2, WN, WD3) forming a landscape-scale woodland-shrubland network. This connectivity enhances habitat resilience and wildlife dispersal potential but increases the sensitivity of the system to fragmentation and hydrological disturbance. Peatland habitats are particularly vulnerable to drainage, compaction and nutrient enrichment; any excavation, groundworks or changes to surface hydrology require careful impact assessment.

Detailed per-polygon habitat data, including threats, listed species and management recommendations, is provided in Appendix [X — Habitat Inventory].

3.3 Flora & Invasive Species

Field Flora Survey

No casual plant species observations were recorded during the field surveys completed to date. However, **vegetation survey data from relevé quadrats is reported separately in Section 3.5 below**, representing a structured botanical dataset distinct from ad-hoc species sightings.

Flora Protection Order Species

No Flora Protection Order (FPO) species listed under the Flora Protection Order 1999 (as amended) were recorded in the desk-based assessment or during preliminary field work. This does not exclude the presence of FPO-protected species (e.g., *Spiranthes spiralis*Autumn Lady's-tresses, *Drosera anglica*Great Sundew) within habitats such as wet grassland, alkaline fens or damp heathland; targeted botanical surveys in June–August may reveal such species if present.

Invasive Non-Native Species

The desk-based assessment identified **seven invasive non-native species**recorded within the broader landscape (1 km buffer and surrounding area) via environmental databases. These species are subject to strict legal controls under EU Regulation 1143/2014 (Invasive Alien Species Regulation) and Irish regulations S.I. 477/2011 and S.I. 374/2024:

Species	Common Name	Risk Category	Irish/EU Status
* <i>Gunnera tinctoria</i> *	Giant-rhubarb	High	EU 1143/2014; S.I. 477/2011; S.I. 374...
* <i>Rhododendron ponticum</i> *	Rhododendron	High	S.I. 477/2011; S.I. 374/2024
* <i>Persicaria wallichii</i> *	Himalayan Knotwe...	Medium–High	EU 1143/2014; S.I. 477/2011; S.I. 374...
* <i>Acer pseudoplatanus</i> *	Sycamore	Medium	Medium Risk IAS (2013)
* <i>Ribes nigrum</i> *	Black Currant	Medium	Medium Risk IAS (2013)
* <i>Potamopyrgus antipodar...</i>	Jenkins' Spire Snail	Medium	Medium Risk IAS (2013)
* <i>Oryctolagus cuniculus</i> *	Rabbit	Medium	Medium Risk IAS (2013)

Legal Obligation: The presence of high-risk species (particularly *Gunnera tinctoria*, *Rhododendron ponticum* and *Persicaria wallichii*) within the 1 km buffer zone indicates a risk of introduction or translocation to the project site during construction or through soil import. These species must not be spread beyond their current distribution under Irish and EU law. An **Invasive Species Survey and Management Plan (ISMP)** is mandatory prior to groundworks (see Section 5 — Recommendations).

Risk Assessment: Habitats with highest colonisation potential include:

- Wet grassland (GS4) and scrub (WS1) — suitable for *Gunnera*, *Rhododendron* and knotweeds in damp conditions
- Immature and semi-natural woodland (WS2, WN) — ideal refuge for *Rhododendron* and *Sycamore*
- Watercourses and riparian zones (FW) — conducive to *Jenkins' Spire Snail* and *Rhododendron*

Pre-Works Requirements: A comprehensive invasive species survey and GIS-based mapping must be completed during April–September prior to site mobil

4. Ecological Constraints

Ecological Constraints

The project site and surrounding landscape present a number of ecological constraints arising from the distribution and sensitivity of habitats, the proximity of European designated sites, and the presence of protected species and invasive non-native species. These constraints are assessed below against standard ecological evaluation criteria (naturalness, rarity, diversity, fragility, size/extent and typicalness) and geographic frame of reference, with recommended mitigation actions structured according to the hierarchy of avoidance, minimisation, remediation and compensation.

Habitat Sensitivity and Evaluation

The site occurs within a landscape dominated by woodland (primarily immature woodland, conifer plantation and semi-natural woodland; c. 2,700 ha across the 1 km buffer), peatland habitats (blanket bog and cutover bog; c. 840 ha) and heath systems (dry siliceous and wet heath; c. 75 ha). Desktop habitat data derived from the National Land Cover Map (2018) have been verified through Phase 1 habitat mapping and relevé surveys

conducted in April–May 2026.

Peatland and Heath Habitats. The principal ecological constraint on site is the presence of extensive upland blanket bog (PB2; 684.95 ha within the broader 1 km zone) and wet heath (HH3; 40.1 ha), both of which are potential analogues for Annex I habitats under the EU Habitats Directive (Council Directive 92/43/EEC). Specifically:

- **Annex I 7130 (Blanket bogs)**— active blanket bog is a priority habitat of high conservation importance for peat carbon storage, hydrological regulation and peatland invertebrate fauna. The mapped PB2 polygons merit field assessment to determine peat depth (>30 cm threshold for Annex I classification), degree of degradation and hydrological integrity. Preliminary desktop evaluation suggests medium–high fragility to drainage, peat extraction and afforestation — all pressures evident in the wider Bellacorick Bog Complex SAC (001922; 3.52 km distant).
- **Annex I 4010 (Northern Atlantic wet heath with *Erica tetralix*)**— the 40.1 ha of mapped wet heath (HH3) represents a potentially sensitive ecosystem dependent on waterlogged peat and oligotrophic conditions. Current conservation status within the nearby Bellacorick Bog Complex is documented as unfavourable (4 of 5 recorded interest features in 'bad' condition; NPWS data). The presence of this habitat type within or immediately adjacent to the site requires verification of species composition (especially presence of *Erica tetralix*, *Erica ciliaris*, *Sphagnum* spp.) and hydrological condition during the botanical field season (May–August).
- **Annex I 4030 (European dry siliceous heath)**— 34.55 ha of dry heath (HH1) is mapped within the buffer zone, characterised by open, heathland vegetation on nutrient-poor, siliceous substrates. This habitat is typified by relatively low shrub diversity (likely dominated by *Calluna vulgaris* and *Erica* spp. based on phytosociological data) but represents a habitat of high naturalness, rarity (restricted distribution in Ireland) and fragility to afforestation, over-grazing and nutrient enrichment.

All three potential Annex I habitats are fragile in response to hydrological change (including lowering of water table through drainage, peat cutting or forestry operations) and require a precautionary approach during design and construction. Given the extensive peatland resource and upland topography, any site-specific disturbance to peat hydrology carries potential for indirect effects on designated sites within the Blacksod-Broadhaven catchment (notably Glenamoy Bog Complex SAC at 1.67 km and Bellacorick Bog Complex SAC at 3.52 km).

Woodland and Scrub. Immature woodland (WS2; 1,507.9 ha) and conifer plantation (WD3; 413.7 ha) dominate the landcover. Immature woodland is typical of afforested peatland or heath and exhibits medium ecological sensitivity, particularly where it retains open ground flora or provides nesting and foraging habitat for protected birds (e.g., Goldcrest, *Regulus regulus*, a BoCCI Amber-listed species with 23 records within the desk study area). Semi-natural woodland (WN; 17.16 ha) is rarer and merits assessment for botanical interest (rare bryophytes, pteridophytes) and bat roost potential. Conifer plantations are generally of lower conservation value but require pre-clearance surveys for nesting birds and bat roosts prior to any felling operations.

Grassland and Marginal Habitats. Wet grassland (GS4; 38.66 ha) is mapped within the buffer and may support botanical interest, particularly if species-rich (e.g., containing sedges, rushes or specialist forbs). A targeted botanical survey during peak flowering (June–August) is recommended to assess whether these grasslands merit classification as semi-natural or otherwise sensitive. Improved grassland (GA1; 0.56 ha) is of low ecological interest.

Proximity to Designated Sites and Natura 2000 Network

Two European sites (SACs) occur within close proximity of the project boundary:

Desig...	**Dista...	**Qualifying Interests	**Cons...	**Constraint Implication**
Glenamo...	1.67 km	Blanket bog (7130), Northern Atl...	Not spe...	Potential hydrological/functional connec...
Bellacori...	3.52 km	Blanket bog (7130), Wet heath (4...	0 favour...	Documented pressures (drainage, fore...

Additionally, multiple National Heritage Areas (NHAs) and proposed NHAs (pNHAs) dominated by peatland habitat occur within 2 km, including Inagh Bog NHA (002391; 1.52 km), Ummerantarry Bog NHA (001570; 1.68 km) and Bellacorick Bog Complex pNHA (within Glenamoy/Bellacorick complex). These sites are legally protected under the Wildlife Act 2000, as amended, and qualify habitats under the EU Habitats Directive.

Water Framework Directive and Catchment Linkages. The site is located within the Blacksod-Broadhaven Water Framework Directive (WFD) management unit (Catchment Code 33; River Basin District: Island of Ireland East-West). The Owenmore (Mayo) River (OWENMORE_010; IE_WE_33O040050) occurs 0.51 km from the site and is a monitored WFD waterbody. Surface-water drainage from the site is therefore likely to discharge into the Owenmore system and onwards to the Blacksod-Broadhaven estuary, which forms part of the Mullet/Blacksod Bay Complex SAC (IE0000470). This estuarine SAC supports qualifying habitats (mudflats 1140, large shallow inlets 1160, natural eutrophic lakes 3150) and the Annex II species Otter (*Lutra lutra*). Deterioration of water quality or hydrological regime at the site could potentially breach WFD objectives and indirectly impact downstream European site integrity. Precise WFD status and risk classification for the Owenmore and Blacksod-Broadhaven catchment was not provided in the dataset; this represents a key data gap and should be obtained from the Environmental Protection Agency prior to final design.

Protected Species

Breeding Birds. Goldcrest (*Regulus regulus*) is recorded within the study area (23 records) and is protected under the Wildlife Acts, with Amber status on the Birds of Conservation Concern

5. Discussion & Conclusions

Synthesis of Ecological Findings

The Preliminary Ecological Appraisal has identified a landscape of significant ecological interest centred on extensive peatland, heath and woodland habitats within and surrounding the study area in north-west Mayo. Field survey work to date has confirmed the presence of **upland blanket bog (PB2: 684.95 ha)**, **cutover bog (PB4: 153.65 ha)**, **wet heath (HH3: 40.1 ha)** and **dry siliceous heath (HH1: 34.55 ha)** alongside substantial areas of **immature woodland (WS2: 1507.9 ha)** and **conifer plantation (WD3: 413.7 ha)**. These habitats are distributed across a mosaic that reflects past land-use change (afforestation, peat extraction) and ongoing ecological transitions.

The two vegetation relevés (REL 101 and REL 102) recorded from peat soils classified as **FL1** habitat indicate **acidic grassland with heather**, dominated by graminoids (*Agrostis capillaris*, *Anthoxanthum odoratum*: 40–55% combined) and *Calluna vulgaris* (heather; 8%). With 90% total vegetation cover, firm peat substrate

and relatively open structure (5% canopy), these plots represent either **degraded peatland margins** or **heath–grassland transitions** of the type commonly associated with peat cutting impacts and grazing pressure. The aquatic survey (conducted April 2026) recorded a river channel of 50 m width, clear water (pH 7, conductivity 50 µS/cm, dissolved oxygen 50 mg/L, bedrock substrate), consistent with oligotrophic upland watercourses typical of blanket bog catchments.

However, critical **data gaps** limit definitive interpretation of habitat quality and Annex I designation. Botanical survey has not yet confirmed whether PB2 areas meet the criteria for **Annex I habitat 7130 (Blanket bogs)** in an active state, nor whether HH3/HH1 qualify as **Annex I 4010 (Northern Atlantic wet heaths)** or **4030 (European dry heaths)**. Field verification of peat depth, hydrological integrity and vegetation composition is essential before assessment against EU Habitats Directive requirements can be finalised.

Potential Impacts on Ecological Receptors

Peatland and Heath Habitats— Any direct habitat loss or hydrological disruption affecting PB2, PB4, HH3 or HH1 could result in:

- **Permanent loss or fragmentation** of potential Annex I peatland/heath habitats, triggering adverse effects on site integrity if European sites share hydrological or hydromorphological linkage;
- **Drainage impacts:** peat oxidation, carbon loss, subsidence and ecosystem function degradation affecting water storage, biodiversity and climate resilience;
- **Cumulative water-quality deterioration** via runoff (sediment, nutrient and hydrocarbon loading) to the Owenmore (Mayo) River (0.51 km distance) and downstream to Blacksod–Broadhaven catchment and associated SAC/coastal habitats.

Watercourses and Aquatic Receptors— The Owenmore (Mayo) River (OWENMORE_010, WFD body IE_WE_330040050) lies within the Blacksod–Broadhaven catchment and is hydrologically linked to **Mullet/Blacksod Bay Complex SAC (IE0000470)**, which supports Annex I habitats (mudflats, shallow bays) and Annex II species (*Lutra lutra*, otter). Groundwater pathways to other listed rivers (Cloonaghmore, Ballinglen, Duvowen) may also exist. Potential construction-phase impacts include:

- **Silt loading and turbidity** from exposed ground, vehicle tracking, concrete works and shallow excavation;
- **Hydrocarbon contamination** from fuel/oil spillage, cement leachate and dust suppression agents;
- **Thermal stress** and **dissolved oxygen depletion** from abstraction, disrupted riparian shade or water impoundment.

Protected and Notable Species— **Goldcrest** (*Regulus regulus*) records (BoCCI Amber-listed, protected under Wildlife Acts 1976–2021) indicate likely breeding populations in surrounding woodlands. Clearance of scrub, treelines or young forest during the nesting season (March–August inclusive) risks direct mortality, nest destruction and breeding failure. **Breeding bird surveys** have not been conducted, leaving abundance and distribution unquantified.

Bats— Mature trees within PB2/HH1/HH3 areas and treelines (WL2: 0.16 ha) present potential summer roost and commuting habitat for woodland-dependent species. Buildings and bridges within the 1 km buffer (BL3: 28.77 ha) may shelter hibernacula or summer roosts. No **Presence/Roost Feature (PRF) assessment** has been undertaken; thus, risk from works (tree felling, building demolition) cannot be quantified.

Invasive Species— The desk assessment identified **high-risk invasive species**(*Gunnera tinctoria*, *Rhododendron ponticum*, *Himalayan knotweed**Persicaria wallichii*) within the extended study area. If present on the development footprint, legal obligations under **EU Regulation 1143/2014** and **Irish S.I. 477/2011 and S.I. 374/2024** require mandatory eradication and stringent biosecurity. Uncontrolled clearance risks spread to adjacent semi-natural habitats and watercourses.

Cumulative and In-Combination Effects

The study area is embedded within a sensitive regional peatland complex of high conservation value.

Bellacorick Bog Complex pNHA/SAC (001922, 3.52 km distance) comprises 9585.6 hectares of blanket bog, wet heath, dystrophic lakes and Rhynchosporion depressions, with current conservation status recorded as poor (0 favourable, 1 inadequate, 4 bad of 5 qualifying interests). Upstream catchment pressures including **drainage, peat extraction, afforestation, erosion and burning** are already documented threats.

Cumulative hydrological effects— Should the proposed development involve:

- Peat extraction, deep drainage or groundwater abstraction;
- Compaction of peat or alteration of microtopography (pools, hummocks);
- Diversion or culverting of drainage lines or flushes;

...then combined with existing catchment disturbances, these could exacerbate:

- **Lowering of water tables** in adjacent bog/heath (accelerating drying and plant community shifts);
- **Reduced base-flow** and **increased peak-flow variability** in watercourses, destabilising aquatic and riparian communities downstream;
- **Eutrophication and acidification** if peat oxidation and drainage-driven nutrient leaching occur, degrading conditions for oligotrophic bog flora and dependent fauna.

In-combination effects with climate change— Blanket peatlands are acutely sensitive to water-table drawdown under projected increases in summer evapotranspiration and periodic drought. Degradation of peat structure, loss of Sphagnum and shift towards grass dominance (as observed in relevés) reduce site resilience to climate variability. Development-driven drainage would amplify climate-driven stress, reducing carbon storage and biodiversity value synergistically.

Invasive species pathway— Disturbance and bare ground created during construction provide vectors for establishment of *Rhododendron*, *Gunnera* and knotweeds, which can spread into adjoining semi-natural habitats and waterside vegetation if not actively managed.

Application of the Precautionary Principle

Given the proximity of **two SACs** (Glenamoy Bog Complex SAC at 1.67 km; Bellacorick Bog Complex SAC at 3.52 km), the presence of potential Annex I habitats within the study area, confirmed hydrological connectivity to the Owenmore River and linked coastal European sites, and **significant data gaps** in botanical, hydrological and species surveys, the precautionary principle requires that

6. Appendices

APPENDICES

Part 1 — Appendix Contents

The following appendices provide detailed supporting data and documentation for this Preliminary Ecological Appraisal:

Appendix A: Site Location Map Plan-form map showing the site boundary (centred on grid reference E 062 293, Shanettra, County Mayo) with a 1 km buffer zone overlaid. Designated sites (SACs, pNHAs, NHAs) within 2 km of the site are marked and labelled with their conservation codes and qualifying interests.

Appendix B: Habitat Map (FOSSITT Classification) Detailed habitat mapping derived from the National Land Cover Map 2018, with all 31 habitat polygons classified according to Fossitt (2000). Habitats are colour-coded by type (peatland, heath, woodland, grassland, artificial surfaces, watercourses) and labelled with their FOSSITT codes, approximate areas in hectares, and condition assessments as derived from field survey (22 April 2026) and desk appraisal.

Appendix C: Species Records and Survey Data Complete dataset of species observations, relevé vegetation surveys (two relevés: REL 101 and REL 102, both classified as FL1 mesotrophic flush or fen grassland, recorded 22 April 2026), and field survey summary sheets. Includes species abundance assessments (DAFOR), protected status flags, and associated fauna observations.

Appendix D: Photographs A selection of site photographs taken during the habitat mapping and relevé surveys of 22 April 2026, annotated with location, habitat type, notable features and any protected or invasive species observed. Photographs document the condition and character of dominant habitats (immature woodland, peatland, heath and grassland).

Appendix E: Survey Datasheets and Field Records Raw field survey records compiled on 22 April 2026 (habitat mapping, relevé surveys, and aquatic assessment) and 9 May 2026 (walkover survey in progress), including weather conditions, surveyor notes, habitat structure measurements, vegetation cover percentages, and faunal observations.

Part 2 — Relevant Data and References

The following data sources were consulted during the preparation of this Preliminary Ecological Appraisal. All web links were accessed between April and May 2026.

- **National Land Cover Map (NLC) 2018.** Tailte Éireann.
<https://www.tailte.ie/surveying/products/professional-mapping/national-land-cover/>
- Source of desktop habitat classification and mapping for all 31 habitat polygons within the site boundary and 1 km buffer zone. Habitats classified according to Fossitt (2000) nomenclature and grouped by sensitivity and conservation interest.
- **National Parks and Wildlife Service (NPWS) Designated Sites Database.** <https://www.npws.ie>
- Bellacorick Bog Complex pNHA (site code 001922): Consulted for qualifying interest habitats (natural dystrophic lakes, northern Atlantic wet heaths with *Erica tetralix*, blanket bogs, Rhynchosporion

depressions, alkaline fens) and conservation status summary.

- Glenamoy Bog Complex (site code 000500): SAC and pNHAdesignations; peatland habitat interest noted but detailed conservation objectives not provided.
- Inagh Bog NHA (site code 002391): Peatland NHA designation; conservation details not available from source.
- Ummerantarry Bog NHA (site code 001570): Peatland NHA; conservation details not provided.
- Mullet/Blacksod Bay Complex SAC (site code IE0000470): Linked to Blacksod-Broadhaven catchment; qualifying intertidal and shallow bay habitats and otter (*Lutra lutra*) as Annex II species.
- **Environmental Protection Agency (EPA) Water Resources.**<https://www.epa.ie>
 - Blacksod-Broadhaven catchment (catchment code 33, River Basin District: Ireland, Eastern and Western): WFD management unit intersecting the project site; WFD status and detailed risk assessment data not available from desktop source.
 - Owenmore (Mayo) River [water body code IE_WE_33O040050]: Located approximately 0.51 km downgradient of the site; potential hydrological connectivity confirmed.
 - Associated river water bodies: CLOONAGHMORE_010 (IE_WE_34C030030), BALLINGLEN_010 (IE_WE_33B010100), DUVOWEN_010 (IE_WE_34D030800) and Moy & Killala Bay catchment (code 34) mapped within the broader catchment context; WFD status not provided.
- **Fossitt, J.A. (2000).***A Guide to Habitats in Ireland.*Heritage Council, Kilkenny.
 - Standard reference for Irish habitat classification and description. Used to classify all 31 habitat polygons and to assess potential Annex I (EU Habitats Directive) correspondences for peatland (codes 7130, 7150), heath (codes 4010, 4030) and fen (code 7230) habitats on and adjacent to the site.
- **Smith, G.F., O'Donoghue, P., O'Hora, K. & Delaney, A. (2011).***Best Practice Guidance for Habitat Survey and Mapping.*Heritage Council, Kilkenny.
 - Methodological reference for Phase I and detailed habitat assessment protocols. Applied in field verification of habitat extent and condition during the walkover survey of 22 April 2026 and ongoing survey work (as of 9 May 2026).
- **CIEEM (2018).***Guidelines for Ecological Impact Assessment in the UK and Ireland. Guidance for Assessing the Significance of Ecological Effects of Proposed Development.*Chartered Institute of Ecology and Environmental Management, Winchester.
 - Principal guidance framework for the scope, methodology and assessment narrative of this PEA. Applied to evaluate ecological sensitivity, data gaps, and recommendations for further survey work and mitigation.
- **Wildlife Acts 1976–2021 (Ireland).**
 - Primary legal framework governing the protection of wild flora and fauna in Ireland. Applied to assess the protected status of species recorded (notably Goldcrest *Regulus regulus*, Amber-listed under Birds of Conservation Concern) and to define exclusion periods for vegetation clearance (March–August for nesting birds).
- **EU Directive 2009/147/EC (Birds Directive) and Directive 92/43/EEC (Habitats Directive).**
 - Framework for the assessment of Annex I habitats and Annex II species with regard to designated European sites (SACs and SPAs). Consulted for conservation objectives and qualifying interests of Bellacorick Bog Complex SAC, Mullet/Blacksod Bay Complex SAC, and Glenamoy Bog Complex

SAC.

- **Regulation (EU) No. 1143/2014 (Invasive Alien Species Regulation) and Irish Statutory Instruments S.I. 477/2011 and S.I. 374/2024.**
 - Legislation governing the control and prohibition of invasive non-native species. Applied to flag high-risk invasive plants identified in desk research (*Gunnera tinctoria*, *Rhododendron ponticum*, Himalayan knotweed *Persicaria wallichii*) and to justify the requirement for a pre-works invasive species survey and management plan.

RELEVÉ DATA — APPENDIX I

Relevé REL 101

Metadata	**Value**
Relevé Code	REL 101
Date Recorded	22 April 2026
Recorder	test
Site Name	Test apro
GPS Coordinates	53.34980 N, 6.26030 W
GPS Accuracy	±5 m
Habitat Type (FOSSITT)	FL1 (Mesotrophic Flush or Fen Grassland)
Soil Type	Peat
Soil Stability	Firm

Appendix A: Habitat Map

[Content to be inserted]

Appendix B: Designated Sites

Name	Site Number	Distance
Inagh Bog NHA	002391 (NHA)	1.5 km
Ummerantary Bog NHA	001570 (NHA)	1.7 km
Ummerantary Bog NHA	001570 (NHA)	1.7 km
Glenamoy Bog Complex	000500 (pNHA)	1.7 km
Glenamoy Bog Complex SAC	000500 (SAC)	1.7 km
Bellacorick Bog Complex SAC	001922 (SAC)	3.5 km

Appendix C: Species List

Name	Protection Status
Regulus r...	Protected Species: Wildlife Acts Threatened Species: Birds of Conservation Concern Threaten...
Acer pse...	Invasive Species: Medium Risk Invasive Species (2013 Report) Invasive
Gunnera t...	Invasive Species: EU Invasive Alien Species Regulation No. 1143/2014 Invasive Species: Regul...
Oryctolag...	Invasive Species: Medium Risk Invasive Species (2013 Report) Invasive
Persicaria...	Invasive Species: EU Invasive Alien Species Regulation No. 1143/2014 Invasive Species: Regul...
Potamopy...	Invasive Species: Medium Risk Invasive Species (2013 Report) Invasive
Rhodode...	Invasive Species: Regulation S.I. 477/2011 (Ireland) Invasive Species: Detailed Risk Assesment ...
Ribes nig...	Invasive Species: Medium Risk Invasive Species (2013 Report) Invasive

Appendix D: Aquatic Features

Name	Type	WFD Status	Distance
OWENMORE (MAYO)_010	River	—	0.5 km
Moy & Killala Bay	Catchment	—	3.3 km
CLOONAGHMORE_010	River	—	3.6 km
BALLINGLEN_010	River	—	4.4 km
DUVOWEN_010	River	—	5.3 km
Blacksod-Broadhaven	Catchment	—	—
Blacksod-Broadhaven	Catchment	—	—

Appendix E: Site Photographs

[Content to be inserted]

Appendix F: Habitat Data

Fossitt Code	Habitat Category	Area (ha)	% Cover
BL3	Buildings and artificial surfaces	—	0.0%
BL3	Buildings and artificial surfaces	0.51 ha	0.0%
BL3	Buildings and artificial surfaces	7.82 ha	0.3%
BL3	Buildings and artificial surfaces	4.79 ha	0.2%
BL3	Buildings and artificial surfaces	15.95 ha	0.5%
ER	Exposed rock	1.43 ha	0.0%
ER	Exposed rock	1.18 ha	0.0%
FW	Watercourses	6.27 ha	0.2%
GA1	Improved agricultural grassland	0.56 ha	0.0%
GS4	Wet grassland	19.47 ha	0.7%
GS4	Wet grassland	19.19 ha	0.6%
HD1	Dense bracken	0.54 ha	0.0%
HH1	Dry siliceous heath	30 ha	1.0%
HH1	Dry siliceous heath	4.55 ha	0.2%
HH3	Wet heath	37.63 ha	1.3%
HH3	Wet heath	2.47 ha	0.1%
PB2	Upland blanket bog	679.78 ha	22.7%
PB2	Upland blanket bog	5.17 ha	0.2%
PB4	Cutover bog	124.23 ha	4.2%
PB4	Cutover bog	17.1 ha	0.6%
PB4	Cutover bog	12.32 ha	0.4%
WD3	Conifer plantation	145.71 ha	4.9%
WD3	Conifer plantation	267.99 ha	9.0%
WL1	Hedgerows	1.55 ha	0.1%
WL2	Treelines	0.16 ha	0.0%
WN	Semi-natural woodland	17 ha	0.6%
WN	Semi-natural woodland	0.16 ha	0.0%
WS1	Scrub	0.69 ha	0.0%

WS1	Scrub	31.2 ha	1.0%
WS2	Immature woodland	735.31 ha	24.6%
WS2	Immature woodland	772.59 ha	25.8%

Appendix G: Survey Datasheets

[Content to be inserted]

Appendix H: Legislation References

[Content to be inserted]